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MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY (AMPLIFIER-LOUDSPEAKER EQUIPMENT, FREQUENCY RANGE 100 TO 7000 CPS)

1. SCOPE

1.1 This specification covers amplifier-loudspeaker equipment for use in enclosed spaces such as wardrooms or messrooms aboard ship. In general, the source of program material will be radio receivers or record players designed for shipboard installation.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power, and Pulse).
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-S-3786 - Switches, Rotary (Circuit Selector, Low Current Capacity), General Specification For.
MIL-S-3950 - Switches, Toggle
MIL-P-15024 - Plates, Identification____Information and Marking for Identification of Electrical, Electronic, and Mechanical Equipment.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-701 - Preferred and Guidance List of Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.2).

3.2 Description.- The amplifier-loudspeaker equipment shall be capable of reproducing the output signal from Navy audio equipment having an output impedance of 600 ohms. The amplifier shall be transistorized and housed within the loudspeaker enclosure. Means of selecting any one of five input channels shall be provided.

FSC 5830

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3.3 General features.- The equipment shall be in accordance with the following paragraphs of MIL-I-983, in addition to the requirements specified herein. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern:

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Magnesium
Iron and steel
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Locking devices
Washers
Parts-electrical-general
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers
Electrical tapes
Switches (see 3.4.3.3)
Indicator lights and lampholders
Fuses
Fuseholders and fuse clips
Metallic rectifiers
Enclosures - general (see 3.5.1)
Enclosure - accessibility
Enclosure - mounting
Shock, vibration (see 3.5.4.2 and 3.5.4.3) and inclination
Ventilation
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (special cold limit applies) (see 3.5.4.1)
Primary power supply circuits (fuses required)
Power supply tolerances (see 3.5.3.2)
Personnel protection
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (test cable required)
Terminal boards, connectors and terminals
Wiring
Color coding
Electrical insulation

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Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams (see 3.4.4.1)
Wiring diagram
Drawing list
Assembly drawings
Interchangeability and standardization
Manuals
Repair parts (electronic)
Designation and marking (serial numbers are required)
Reports
Item names and nomenclature
Workmanship and general examination

3.4 General design.-

3.4.1 The amplifier-loudspeaker shall consist of the following major items:

(a) Enclosure.
(b) Loudspeaker.
(c) Amplifier.

The loudspeaker and enclosure are treated as a unit in the acoustical requirements and test section of this specification. The transistor amplifier is treated as a separate unit.

3.4.2 Dimensions and weight.- The dimensions and weight of the equipment shall not exceed the following limits:

Height ---------- 16 inches.
Width ---------- 14 inches.
Depth ---------- 10 inches.
Weight ---------- 20 pounds.

3.4.3 Component parts.- All electrical and electronic components used in the equipment which are not Military approved types shall be specifically approved by the bureau or agency concerned. Approval will only be granted where no Military approved type exists for the specific application.

3.4.3.1 Transistors.- All transistors shall conform to MIL-STD-701. No electron tubes shall be used in the equipment.

3.4.3.2 Transformers.- Transformers shall be grade 4 or 5, class S, life expectancy X of MIL-T-27.

3.4.3.3 Switches.- Switches shall be in accordance with MIL-S-3786 or MIL-S-3950.

3.4.4 Identification plate.- A type A or B identification plate in accordance with MIL-P-15024 shall be mounted on the front of the enclosure. Equipment and component line shall read "Amplifier-Loudspeaker Equipment".

3.4.4.1 Schematic diagram.- A type B or H plate in accordance with MIL-P-15024 shall be securely mounted within the enclosure where it will be easily readable with the cover open. This plate shall include a schematic diagram of the entire electronic system showing connections to all incoming cables and a parts list.

3.5 Physical requirements.- The amplifier-loudspeaker shall withstand long exposure to high humidity, operating temperature extremes, vibration, shock and other conditions encountered in shipboard service.

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3.5.1 Enclosure.- The enclosure shall be constructed of metal or of fibrous glass reinforced polyester plastic and shall be designed with three 1/2 inch diameter reinforced holes in the back plate for bulkhead mounting. The enclosure shall have an opening for the loudspeaker face. This opening and any port openings required to meet the specified performance characteristics, shall be covered with metal mesh or perforated metal not less than 0.040 inch thick. Mesh openings shall be not larger than 1/16 inch. Metal covering such openings shall be anodized aluminum or corrosion-resistant steel, painted in accordance with MIL-I-983. The enclosure shall be provided with a hinged front cover opening downward 180 degrees. The cover shall be held closed by a spring type catch (or catches) requiring no tools for operation. Opening the cover shall provide ready access to all interior components and to the enclosure mounting holes. Opening the cover shall not involve disturbance of incoming cables and terminal tubes. Component mounting fasteners shall be secured so as not to require demounting of the enclosure for component replacement. Components shall be mounted so that their terminal markings are easily visible with the cover open. Damping material if used shall not disintegrate or become dislodged during shock and vibration tests herein specified. Facilities shall be provided to admit cables through the top or bottom of the enclosure utilizing stuffing tubes as specified in MIL-I-983. A terminal board shall be mounted on the inside surface of the back of the enclosure for connection of all incoming cables. Painted surfaces shall be within the color limits allowed in MIL-I-983.

3.5.2 Loudspeaker.- The loudspeaker shall be a nominal 8 inch diameter direct radiator type. The method of mounting shall be in accordance with commercial standards.

3.5.3 Amplifier.- The transistor amplifier shall be mounted in the lower portion of the enclosure and shall be designed with plug in connections to permit easy removal and replacement.

3.5.3.1 Amplifier controls.- The following controls shall be located on the hinged front cover:

(a) Alternating current (a.c.) power switch.- The power switch shall be a two pole, two position toggle switch which shall connect and disconnect each side of the a.c. power line from the amplifier power supply.
(b) Pilot light.- The pilot light shall provide a visible indication that the equipment has been energized. A mechanical means of dimming the light shall be provided.
(c) Line fuses.- Fuses shall be installed in each side of the a.c. power line between the power switch and the power transformer. The fuse-holders shall be equipped with blown fuse indicating lamps. The fuse rating shall be marked at each fuse.
(d) Channel selector switch.- The channel selector switch, titled "CHANNEL SELECTOR", shall provide a means of selecting any one of five input channels. The switch shall be a rotary selector switch as specified in MIL-S-3786. The five positions shall be marked 1, 2, 3, 4, and 5 in clockwise sequence.
(e) Volume control.- The volume control, titled "VOLUME" shall be a potentiometer provided with a scale divided into ten equal divisions and marked from 0 to 10 in clockwise sequence. The characteristic of the volume control shall provide linear attenuation in decibels within six decibels (db) over approximately 75 percent of the shaft rotation. Maximum attenuation shall occure at full counterclockwise rotation and shall be at least 50 db down from setting which produce rated output with 1.0 volt input.

3.5.3.2 Amplifier power supply requirements.- The amplifier power supply shall be designed to operate from a single phase a.c. power source of 115 volts, 60 cycles. The power supply shall consist of a power transformer, rectifier, and filter. Silicon power diodes shall be used for the rectifier. Plug-in electrolytic filter capacitors shall be permitted if they are provided with suitable clamping devices.

3.5.3.3 Supply line voltage and frequency extremes.- The amplifier shall be capable of satisfactory operation when energized from power sources whose operating voltage and frequency may vary between the limits of 115 ± 7 volts and 60 ± 5 cycles per second (c.p.s.), respectively. No evidences of overheating, instability or other undesirable effects shall occur when operated at the specified extremes for extended periods of time.

3.5.4 Environmental requirements.-

3.5.4.1 Temperature and humidity.- The equipment shall be capable of satisfactory operation when subjected to ambient temperatures between the limits of 40° and 131° Fahrenheit (F.) and in any relative humidity up to 95 percent. Attention is called to the requirement for design adequate to withstand extended storage at ambient temperatures between minus 80° and plus 167° F.

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3.5.4.2 Vibration.- The equipment shall operate satisfactorily after being subjected to the vibration test for "nonvital" equipment specified in MIL-I-983.

3.5.4.3 Shock.- The equipment shall withstand the shock test of MIL-S-901 without the enclosure tearing loose from the mounting.

3.6 Performance requirements.-

3.6.1 Amplifier performance.- The amplifier performance characteristics shall be considered independently of the loudspeaker, that is, an 8 ohm, 10 watt resistance shall be substituted for the loudspeaker as the load on the amplifier. This load shall be considered to be the rated load of the amplifier.

3.6.1.1 Rated operating conditions.- The amplifier performance characteristics except as otherwise specified shall be determined at rated line voltage and frequency of 115 volts, 60 cycles. The room ambient temperature shall be 77° ± 9°F. (25° ± 5° Centigrade (C.)). The relative humidity shall not exceed 60 percent.

3.6.1.2 Stability.- The amplifier shall not exhibit any evidence of instability such as drift in gain or spurious oscillations when energized at any line voltage within the line voltage and frequency limits specified in 3.5.3.3 nor shall there be evidence of instability with input circuit short circuited or open-circuited.

3.6.1.3 Amplifier performance characteristics.- The amplifier performance characteristics as determined under rated operating conditions (see 3.6.1.1) shall be within the requirements shown in table I. The method of measurement shall be as specified in table I.

Table I - Amplifier performance characteristics.

Characteristic	Performance requirement	Method of measurement paragraph
Input impedance	6000 ± 600 ohms	4.6.3.1
Frequency response	Within limits of 2 db (+1 db)	4.6.3.2
Input signal level range	0.5 to 35 volts	4.6.3.2 and 4.6.3.4
Signal-to-noise ratio	60 db	4.6.3.3
Distortion	3 percent	4.6.3.4
Rated output power	5.0 watts into 8 ohm resistive load	4.6.3.4

3.6.2 Loudspeaker performance.- The loudspeaker enclosure shall meet the performance requirements shown in table II. The individual performance characteristics shall, in general, be determined in the order indicated.

Table II - Loudspeaker performance characteristics.

Loudspeaker characteristic	Performance requirement	Method of measurement paragraph
Endurance	24 hours	4.6.4.2
Impedance	8 ± 1 ohms at 400 c.p.s.	4.6.4.3
Output versus input	Linear within 1 db	4.6.4.4
Frequency response	Within limits shown on figure 1	4.6.4.5
Output sound pressure	94 db (ref. 0.0002 dyne/cm. ^{2})	4.6.4.6

3.7 Test cable.- One test cable subassembly shall be provided as specified in MIL-I-983 to permit operation and test of the equipment with the amplifier removed from the enclosure.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection.- The procedures for the various tests to be performed on the equipment shall be as specified hereinafter. (The attention of the manufacturer is invited to the applicable paragraphs of MIL-I-983 for information concerning these tests such as where conducted, failures, and test conditions.)

4.3 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in table III, which shall be performed in general in the order listed. These tests shall be conducted at a laboratory satisfactory to the Bureau of Ships:

Table III - Preproduction inspection.

Examination and tests	Requirement paragraph	Examination and test paragraph
General examination	MIL-I-983	4.6.1
Operating tests	3.5.3.1	4.6.2
Amplifier measurements:	3.6.1.3	4.6.3
Input impedance	3.6.1.3	4.6.3.1
Frequency response	3.6.1.3	4.6.3.2
Input signal level range	3.6.1.3	4.6.3.2 and 4.6.3.4
Signal-to-noise ratio	3.6.1.3	4.6.3.3
Distortion	3.6.1.3	4.6.3.4
Rated output power	3.6.1.3	4.6.3.4
Stability	3.6.1.2	4.6.3.5
Loudspeaker measurements:	3.6.2	4.6.4 and 4.6.4.1
Endurance	3.6.2	4.6.4.2
Impedance	3.6.2	4.6.4.3
Output versus input	3.6.2	4.6.4.4
Frequency response	3.6.2	4.6.4.5
Output sound pressure	3.6.2	4.6.4.6
Supply line voltage and frequency extremes	3.5.3.3	MIL-I-983
Temperature and humidity	3.3 and 3.5.4.1	MIL-I-983
Vibration	3.3 and 3.5.4.2	MIL-I-983
Shock	3.3 and 3.5.4.3	MIL-S-901

The result of each test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure and the entire series of examination and tests repeated. Where the corrections made can be shown to have made no change in the ability to meet a specific unrelated examination and test, this examination and test may be waived at the discretion of the Government.

4.4 Sampling for quality conformance inspection.-

4.4.1 Lot.- All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.4.2 Group A.- All equipment covered on the contract or order shall be subjected to the group A examination and tests listed in 4.5. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any examination and test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

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4.4.3 Sampling for group B tests.- A sample of equipments shall be selected from each lot in accordance with table IV and shall be subjected to each of the group B tests listed in 4.5 with quality conformance based on sampling requirements of MIL-STD-105. The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any test shall be counted as a defect and the equipment shall not be offered for delivery. If the number of nonconforming equipments in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

Table IV - Sampling for group B tests.

Number of equipments in lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.4.4 Sampling for group C tests.- Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group C tests listed in 4.5. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.5 Quality conformance inspection.- The sample equipment selected in accordance with 4.4 shall be subjected to the examination, and tests listed in table V. Examination and tests shall be performed, in general, in the order listed.

Table V - Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Group A examination and tests		
General examination	MIL-I-983	4.6.1
Operating	3.5.3.1	4.6.2
Group B tests		
Amplifier measurements:	3.6.1.3	4.6.3
Frequency response	3.6.1.3	4.6.3.2
Input signal level range	3.6.1.3	4.6.3.2 and 4.6.3.4
Signal-to-noise ratio	3.6.1.3	4.6.3.3
Distortion	3.6.1.3	4.6.3.4
Rated output power	3.6.1.3	4.6.3.4
Loudspeaker measurements:	3.6.2	4.6.4
Frequency response	3.6.2	4.6.4.5

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Table V - Quality conformance inspection. (cont'd.)

Inspection	Requirement paragraph	Test paragraph
Group C tests		
Amplifier measurements:	3.6.1.3	4.6.3
Input impedance	3.6.1.3	4.6.3.1
Stability	3.6.1.2	4.6.3.5
Loudspeaker measurements:	3.6.2	4.6.4
Endurance	3.6.2	4.6.4.2
Impedance	3.6.2	4.6.4.3
Output versus input	3.6.2	4.6.4.4
Output sound pressure	3.6.2	4.6.4.6
Supply line voltage and frequency extremes	3.5.3.3	MIL-I-983
Temperature and humidity	3.3 and 3.5.4.1	MIL-I-983
Vibration	3.3 and 3.5.4.2	MIL-I-983
Shock	3.3 and 3.5.4.3	MIL-S-901

4.6 Inspection.-

4.6.1 General examination.- The general examination shall be conducted as outlined in MIL-I-983.

4.6.2 Operating tests.- The operating tests shall consist of determining that the complete amplifier-loudspeaker equipment functions as required by this specification. A program of music with 0.5 volt peaks shall be applied to each amplifier input. The volume control shall be adjusted to produce electrical peaks of 6.4 volts at the loudspeaker terminals. There shall be no extraneous or objectionable noises in the loudspeaker output during the above test.

4.6.3 Amplifier measurements.- The methods of measurements by means of which the amplifier performance characteristics specified in 3.6.1.3 shall be determined, are described in 4.6.3.1 through 4.6.3.5. Measurements shall be made at rated operating conditions (see 3.6.1.1) unless otherwise specified.

4.6.3.1 Input impedance.- The input impedance measurements shall be made with the amplifier energized and with the volume control adjusted to delivery rated output power at the center frequency of 1000 c.p.s. The input impedance shall be measured at frequencies of 200, 1000 and 5000 c.p.s. and at input voltage levels of 1.0, 10 and 35 volts. A suitable impedance recorder may be used to make these measurements. The input impedance shall be essentially constant within the limits specified in table I over the frequency range and input signal level range of the amplifier.

4.6.3.2 Frequency response.- With an input of 0.5 volt to the amplifier, the gain control shall be adjusted to deliver 2.5 watts into an 8 ohm resistive load at 1000 c.p.s. With the input voltage held constant at 0.5 volt the frequency shall be swept over a range of 100 to 7000 c.p.s. and the response characteristic determined for conformance with 3.6.1.3. The above test shall be repeated with input voltages of 7.0 and 35.0 volts to determine conformance with 3.6.1.3

4.6.3.3 Signal-to-noise ratio.- The gain control shall be set so that with an input of 0.5 volt at 1000 c.p.s., the output of the amplifier into a resistive load of 8 ohms is 5 watts. With no signal applied to the input and a 600 ohm resistor connected across the input terminals, the voltage output of the amplifier across the 8 ohm resistive load shall be measured. The ratio of this voltage to 6.4 volts shall be computed to determine conformance with 3.6.1.3.

4.6.3.4 Distortion.- With an input of 0.5 volt to the amplifier, the gain control shall be successively adjusted to deliver 5 watts into an 8 ohm resistive load at 200, 400, 1000, and 3000 c.p.s. and to deliver 2.5 watts into an 8 ohm resistive load at 100 and 7000 c.p.s. The distortion content of the amplifier output shall be measured at each frequency to determine conformance with 3.6.1.3. The above test shall be repeated with input voltages of 7.0 volts and 35.0 volts and shall meet the requirements of 3.6.1.3.

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4.6.3.5 Stability. - The amplifier shall be operated at all gain settings with the amplifier input either short or open circuited and with the amplifier output connected to the loudspeaker (in its enclosure) to determine conformance with 3.6.1.2.

4.6.4 Loudspeaker measurements. - The methods of measurements, by means of which the loudspeaker performance characteristics specified in 3.6.2 shall be determined, are described in the following paragraphs: All measurements shall be made at applicable rated operating conditions unless otherwise specified, with the loudspeaker and amplifier installed in its cabinet. The amplifier contained in the unit shall not be used for these measurements. An external amplifier of at least 20 watts capacity, low distortion and low noise shall be employed. Prior to performing any measurements, the loudspeaker shall be subjected to the endurance test of 4.6.4.2 in order to condition the suspension system in addition to determining adequate physical power handling capacity.

4.6.4.1 Acoustic environment. - All acoustic measurements shall be performed with the loudspeaker and measuring microphone located in an essentially free field.

4.6.4.2 Endurance. - A 1000 c.p.s., 9 volt signal interrupted at a rate of between 100 and 115 times per second (t.p.s.) in equal on and off periods (equivalent to an average electrical input of 5 volt amperes) shall be applied to the loudspeaker voice coil continuously for a period of 24 hours. At the conclusion of the endurance test the loudspeaker shall meet all performance requirements of this specification and shall show no evidence of physical damage.

4.6.4.3 Impedance. - The loudspeaker impedance shall be determined with the loudspeaker mounted in its enclosure. A constant current of 10 milliamperes shall be applied to the voice coil through a series resistance of at least 1000 ohms at 400 c.p.s. The voice coil voltage shall be measured and the impedance determined.

4.6.4.4 Output versus input. - An electrical signal warbled in frequency, linearly with time, over the range from 800 to 1250 c.p.s. at a warble rate of approximately 6 c.p.s. shall be applied to the loudspeaker voice coil. The level of the signal shall be increased in steps up to at least 5 voltamperes (6.4 volts) input. At each step the acoustic output pressure shall be measured at 10 feet from the frontmost plane of the loudspeaker cabinet on the axis through the voice coil of the loudspeaker. The signal shall be applied for a sufficient time at each step to permit the voice coil temperature to reach equilibrium before the acoustic output is measured. The acoustic output shall be plotted against input voltage. In addition to the curve thus obtained a straight line shall be drawn between the 0 and the 6.4 volt input point. Deviation from linearity shall be the maximum percentage deviation of the sound pressure from the straight line characteristic at any point below the 6.4 volt input point.

4.6.4.5 Frequency response. - A signal at a constant amplitude of 6.4 volts shall be applied to the loudspeaker voice coil and the frequency swept through a range of 100 to 7000 c.p.s. The output sound pressure of the loudspeaker shall be measured at a distance of 10 feet from the face of the loudspeaker enclosure and on the axis of the loudspeaker cone. The response characteristic shall be in accordance with 3.6.2.

4.6.4.6 Output sound pressure. - The output sound pressure shall be determined from the frequency response characteristic obtained in 4.6.4.5. The response characteristic of the loudspeaker shall be superimposed on and within the curve of figure 1 in such a manner that the lower limit of the curve of figure 1 and the response characteristic touch. This point of touching establishes pressure levels for figure 1. The output sound pressure of the loudspeaker is considered to be that pressure represented by the midpoint between the limits shown on figure 1 at 1000 c.p.s. The output sound pressure shall be as specified in 3.6.2.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Amplifier - loudspeaker equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

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5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- The onboard repair parts shall be preserved and packaged by level A; packed by level C and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- Where special requirements not covered by 5.1 above are required, the levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking.- The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by level A or B as specified and marked in accordance with MIL-E-17555. Level A packaging shall be accomplished by method 1A of MIL-P-116.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Quantity of manuals required (see 3.3).
(d) Inventory control point for repairs parts.
(e) Quantity of repair parts (see 3.3).
(f) Type and quantity of drawings required (see 3.3).
(g) Whether unit stock number is to be entered on the identification plate.
(h) Preservation and packaging, packing, and marking if other than as specified in 5.1 (see 5.2).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5830-N055Sh)

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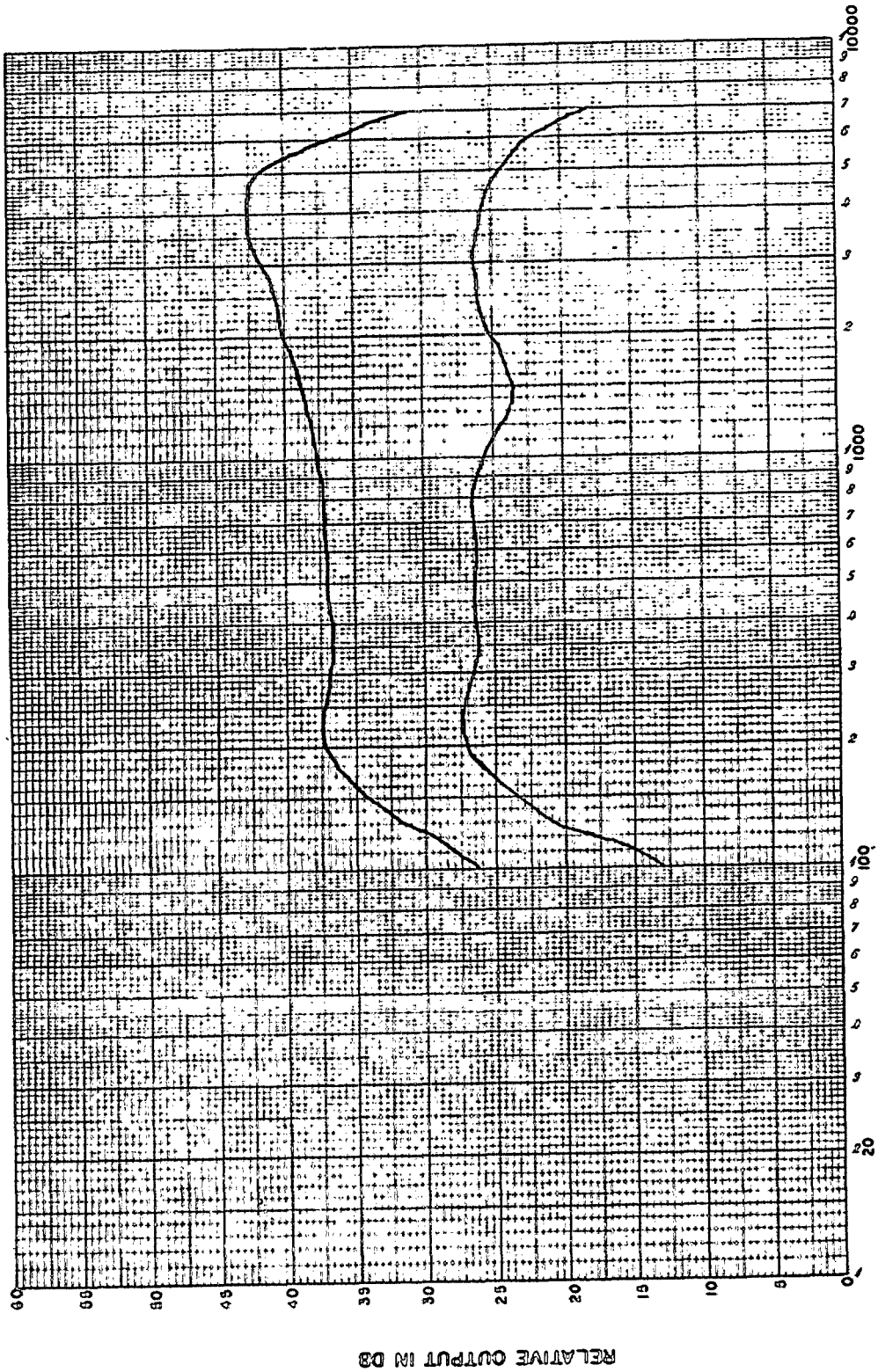


Figure 1 - Frequency response.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R3D9

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION:

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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